BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

BOTANY

SECTION 'A' MULTIPLE CHOICE QUESTIONS (18 Marks)

Maximum Marks: 18

Note: (Options in the original are separated by star marks, shown here as A–D in printed order. Board typos kept verbatim (e.g. 'Hterosporous','Saleginella','paints' for 'plants'). Scan check-marks/ underlines on correct options are not transcribed.)

Note: Time: 2 Hours (Regular & Private) Max. Marks: 45

QQ.1. Choose the correct answer for each from the given options:

Q1. The ability of a microscope to increase the size of an image is:

(A) Resolution
(B) Magnification
(C) Contrast
(D) Objective

Q2. Borderline link between livings and non-livings is:

(A) Bacteria
(B) Fungi
(C) Protozoans
(D) Virus

Q3. It is a Carcinogenic (cancer-causing) mycotoxin:

(A) Alphatoxin
(B) Aflatoxin
(C) Haemolysin
(D) Chitin

Q4. Vascular plants are classified under:

(A) Bacteria
(B) Algae
(C) Bryophytes
(D) Tracheophytes

Q5. It is a Hterosporous plant:

(A) Lycopodium
(B) Fern
(C) Saleginella
(D) Marchantia

Q6. Number of ATPs produced by the complete oxidation of a glucose molecule is:

(A) 36
(B) 38
(C) 37
(D) 40

Q7. This compound connects glycolysis and Krebs cycle:

(A) Pyruvic acid
(B) Acetyle coenzyme A
(C) Citric acid
(D) Oxaloacetate

Q8. Photosynthetic Bacteria use this instead of water:

(A) H2SO4
(B) HCl
(C) H2S
(D) NH3

Q9. It is a wasteful process in C3 paints which decreases crop yield:

(A) Photosynthesis
(B) Respiration
(C) Photo-Respiration
(D) Transpiration

Q10. Excess water is released at night in the form of droplets through:

(A) Stomata
(B) Cuticle
(C) Hydathodes
(D) Lenticels

Q11. During this stage the bacteria grow and multiply rapidly:

(A) LAG phase
(B) LOG phase
(C) Stationary phase
(D) Decline phase

Q12. This enzyme can fix both CO2 and O2:

(A) Deoxyribose
(B) RUBISCO
(C) PEPG
(D) PEPIS

Q13. Vexillary aestivation is a characteristic of this family:

(A) Fabaceae
(B) Caesalpinacea
(C) Mimosacea
(D) Solanaceae

Q14. Insectivorous plants trap insects to obtain:

(A) Lipid
(B) Enzymes
(C) Cellulose
(D) Nitrogenous Compound

Q15. Process which decomposes pyruvate into three molecules of CO2 is:

(A) C3 cycle
(B) C4 cycle
(C) Kreb's cycle
(D) Benson and Calvin cycle

Q16. The beak-like out growth of ascogenia is called:

(A) Antheridium
(B) Trichogyne
(C) Hyphae
(D) Mycelium

Q17. The miniature bisexual gametophyte of fern is called:

(A) Prothallus
(B) Gametophyte
(C) Sporophyte
(D) Seed

Q18. A classification unit of inbreeding organism is:

(A) Specie
(B) Genus
(C) Family
(D) Phylum

ZOOLOGY

SECTION 'A' MULTIPLE CHOICE QUESTIONS (16 Marks)

Maximum Marks: 16

Note: (Options in the original are separated by star marks, shown here as A–D in printed order. Board typos kept verbatim (e.g. 'inscissors','synthesize'). Q15 4th option obscured — marked. Scan check-marks/underlines on correct options are not transcribed.)

Note: Time: 2 Hours (Regular & Private) Max. Marks: 40

QQ.1. Choose the correct answer for each from the given options:

Q1. The largest invertebrate of the world is:

(A) Whale
(B) Elephant
(C) Giraffe
(D) Squid

Q2. Amoebiasis is caused by:

(A) Virus
(B) Entamoeba gingavalis
(C) Fungi
(D) Entamoeba histolytica

Q3. Water vascular system is found in:

(A) Cockroach
(B) Starfish
(C) Earthworm
(D) Sponge

Q4. The air always present in the human lungs is called:

(A) Tidal Volume
(B) Vital Volume
(C) Residual Volume
(D) Total Lung Capacity

Q5. Human sex hormones are synthesize by this:

(A) Protein
(B) Triglyceride
(C) Steroid
(D) Carbohydrate

Q6. Antibodies are produced by:

(A) B Cells
(B) T Cells
(C) Macrophages
(D) Neutrophils

Q7. RBCs are produced in:

(A) Kidney
(B) Heart
(C) Brain
(D) Bone Marrow

Q8. Oldest Fossil bird of the world is:

(A) Archaeopteryx
(B) Platipus
(C) Hedge Hog
(D) Echidna

Q9. Respiratory pigment in muscles is:

(A) Globin
(B) Haemoglobin
(C) Haemocyanin
(D) Myoglobin

Q10. The non-protein part of enzyme is:

(A) Apoenzyme
(B) Holoenzyme
(C) Prosthetic group
(D) Ribozyme

Q11. Nematocytes are present in:

(A) Ascon
(B) Sycon
(C) Leucon
(D) Hydra

Q12. The study of parasites is called:

(A) Parasitology
(B) Ecology
(C) Cytology
(D) Immunology

Q13. Digestion takes place by:

(A) Acids
(B) Alkaline solutions
(C) Enzymes
(D) Minerals

Q14. Coenzyme consists of:

(A) Protein
(B) Carbohydrates
(C) Lipid
(D) Vitamin

Q15. Enzymes of human body are more active at:

(A) 40°C
(B) 37°F
(C) 37°C
(D) [illegible in source scan]

Q16. The number of inscissors in man is:

(A) 2
(B) 4
(C) 6
(D) 8